Product Design and Manufacturing

Assignment Week-9

1. What does the term ASTM refer to in the context of rapid prototyping?
   a) Association of System Test Manufacturers
   b) American Society for Testing and Materials
   c) Advanced System Techniques for Manufacturing
   d) Additive Standardization and Testing Methodologies

2. Wireframe modeling is commonly used in two-dimensional mode in organizations.
   a) True
   b) False

3. During which phase is the 2D slice information generation carried out in Rapid Prototyping?
   a) CAD model
   b) File manipulation
   c) Machine setup
   d) Part building

4. What is the primary output of Rapid Prototyping?
   a) Blueprints
   b) Prototypes or Models
   c) 2D Model
   d) Test plans

5. Which framework utilizes a 'bend welding robot' to store material on a stage as a basic shape?
   a) FDM
   b) MJM
   c) BPM
   d) 3DW

6. What is the primary mechanism behind stereolithography in Rapid Prototyping?
   a) Mechanical milling
   b) UV light-induced polymerization
   c) Injection molding
   d) Thermal sintering

7. SLS heats the entire powder bed to just below the material's melting point before laser modulation.
   a) True
   b) False

8. What is the primary method for joining powder grains together in processes involving discrete particles?
   a) Fusion by heat
   b) Selective Laser Sintering
   c) Gas Phase Deposition
   d) Laser Engineering Net Shaping

9. Paper Lamination Technology (PLT) utilizes a CO_2 laser to cut the cross-sectional shapes of the part.
   a) True
   b) False

10. In Solid Foil Polymerization (SFP), the part is developed utilizing _____.
   a) argon gas
   b) filaments
   c) semi-polymerized foils
   d) All of these